

AI & Machine Learning Terms Glossary

A reference glossary of core artificial intelligence and machine learning terminology, covering foundational concepts, neural networks, natural language processing, large language models, training and evaluation, MLOps, and AI ethics.

7 sections · 102 terms · sample reference document

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Foundational Machine Learning Concepts

Artificial Intelligence (AI)

The broad field of computer science focused on building systems that perform tasks normally requiring human intelligence, such as reasoning, perception, language understanding, and decision-making.

Machine Learning (ML)

A subfield of AI in which systems improve their performance on a task by learning patterns from data, rather than following explicitly programmed rules.

Deep Learning

A subfield of machine learning that uses neural networks with many layers (hence 'deep') to automatically learn hierarchical representations of data, such as images, text, or audio.

Supervised Learning

A machine learning approach where a model is trained on labeled data, learning to map inputs to known correct outputs, such as classifying emails as spam or not spam.

Unsupervised Learning

A machine learning approach where a model finds patterns or structure in data without labeled examples, such as clustering similar customers together.

Reinforcement Learning (RL)

A machine learning approach where an agent learns to make decisions by taking actions in an environment and receiving rewards or penalties based on the outcomes.

Self-Supervised Learning

A training approach where a model generates its own labels from unlabeled data, for example by predicting a masked word from surrounding context, allowing large-scale training without manual annotation.

Feature

An individual measurable property or input variable used by a machine learning model, such as a pixel value in an image or a word in a sentence.

Feature Engineering

The process of selecting, transforming, or creating input variables to improve a model's ability to learn patterns from data.

Label

The correct output or target value associated with a training example in a supervised learning dataset.

Training Set

The portion of a dataset used to fit a model's parameters during training.

Validation Set

A held-out portion of a dataset used during development to tune hyperparameters and check for overfitting, without touching the final test set.

Test Set

A held-out portion of a dataset used only after training is complete, to give an unbiased estimate of how a model will perform on new, unseen data.

Overfitting

A modeling problem where a model learns the training data too closely, including its noise, and as a result performs poorly on new, unseen data.

Underfitting

A modeling problem where a model is too simple to capture the underlying patterns in the data, resulting in poor performance on both training and new data.

Bias-Variance Tradeoff

The balance between a model being too simple to capture patterns (high bias) and too sensitive to fluctuations in training data (high variance); reducing one often increases the other.

Regularization

A set of techniques used to prevent overfitting by discouraging a model from becoming too complex, often by penalizing large parameter values.

Hyperparameter

A configuration setting for a model or training process, such as the learning rate or number of layers, that is chosen before training rather than learned from data.

Cross-Validation

A technique for evaluating a model by splitting the data into multiple folds, training on some folds and testing on the remaining fold, then rotating which fold is held out.

Ensemble Learning

A technique that combines predictions from multiple models to produce a result that is typically more accurate and robust than any single model alone.

Neural Networks & Deep Learning

Neural Network

A computational model loosely inspired by biological brains, consisting of layers of interconnected nodes ('neurons') that transform input data through weighted connections to produce an output.

Artificial Neuron

The basic computational unit of a neural network, which takes weighted inputs, sums them, applies an activation function, and passes the result forward.

Weights

The learnable numerical parameters in a neural network that determine the strength of the connection between neurons, adjusted during training to reduce error.

Bias (Neural Network Term)

An additional learnable parameter added to a neuron's weighted input sum, allowing the model to shift its activation function and fit data more flexibly.

Activation Function

A mathematical function applied to a neuron's output that introduces non-linearity, enabling neural networks to model complex relationships. Common examples include ReLU, sigmoid, and tanh.

Backpropagation

The core algorithm used to train neural networks, which calculates how much each weight contributed to the prediction error and propagates that information backward through the network to update weights.

Gradient Descent

An optimization algorithm that iteratively adjusts a model's parameters in the direction that reduces the loss function, moving 'downhill' toward better performance.

Learning Rate

A hyperparameter that controls how large a step gradient descent takes when updating model parameters; too high can cause instability, too low can make training very slow.

Loss Function

A function that measures the difference between a model's predictions and the actual target values; training aims to minimize this value.

Convolutional Neural Network (CNN)

A type of neural network architecture that uses convolutional layers to detect spatial patterns, widely used for image recognition and computer vision tasks.

Recurrent Neural Network (RNN)

A type of neural network architecture designed for sequential data, where connections loop back so information from previous steps influences later steps.

Long Short-Term Memory (LSTM)

A type of recurrent neural network architecture designed to better capture long-range dependencies in sequences by using gating mechanisms to control what information is retained or forgotten.

Transformer

A neural network architecture based on self-attention mechanisms rather than recurrence, allowing it to process sequences in parallel; it underlies most modern large language models.

Attention Mechanism

A technique that allows a neural network to weigh the importance of different parts of the input when producing an output, rather than treating all inputs equally.

Self-Attention

A form of attention in which a model relates different positions within a single sequence to each other, letting each element attend to every other element to build contextual representations.

Dropout

A regularization technique in which randomly selected neurons are temporarily ignored during training, reducing overfitting by preventing the network from relying too heavily on specific neurons.

Batch Normalization

A technique that normalizes the inputs to a layer within a neural network during training, which can speed up training and improve stability.

Epoch

One complete pass through the entire training dataset during the training of a model.

Batch Size

The number of training examples processed together before a model's parameters are updated once.

Embedding

A learned, typically dense, vector representation of an item such as a word, image, or user, positioned in a continuous space so that similar items end up close together.

Natural Language Processing & Large Language Models

Natural Language Processing (NLP)

The field of AI concerned with enabling computers to understand, interpret, and generate human language.

Large Language Model (LLM)

A neural network, typically transformer-based, trained on massive amounts of text data to predict and generate language, capable of tasks like answering questions, summarizing, and reasoning.

Tokenization

The process of breaking text into smaller units called tokens, such as words, subwords, or characters, which a language model processes as its basic input units.

Token

A discrete unit of text, such as a word or part of a word, that a language model reads or generates as a single step.

Context Window

The maximum number of tokens a language model can consider at once when generating a response, including both the input prompt and the output.

Prompt

The text input given to a language model to elicit a response, which can include instructions, questions, or examples.

Prompt Engineering

The practice of carefully designing prompts to guide a language model toward producing more accurate, relevant, or useful outputs.

Fine-Tuning

The process of further training a pre-trained model on a smaller, task-specific dataset to adapt its behavior for a particular use case.

Pre-Training

The initial phase of training a large model on a broad, general-purpose dataset before it is adapted for specific tasks.

Zero-Shot Learning

A model's ability to perform a task it was never explicitly trained on, using only a natural language description or instruction.

Few-Shot Learning

A technique where a model is given a small number of examples within the prompt to help it understand and perform a new task without additional training.

Chain-of-Thought Prompting

A prompting technique that encourages a language model to generate intermediate reasoning steps before arriving at a final answer, often improving accuracy on complex tasks.

Hallucination

A term describing when a language model generates text that is fluent and plausible-sounding but factually incorrect or not grounded in the provided source material.

Retrieval-Augmented Generation (RAG)

A technique that combines a language model with an external retrieval system, fetching relevant documents or passages at query time and providing them as context so the model can generate more accurate, grounded responses.

Vector Database

A specialized database designed to store and efficiently search high-dimensional vector embeddings, commonly used to power similarity search in retrieval systems.

Semantic Search

A search technique that matches queries to results based on meaning and context, typically using vector embeddings, rather than relying only on exact keyword matches.

Chunking

The process of splitting a large document into smaller segments before embedding and indexing it, so that a retrieval system can return focused, relevant passages.

Reranking

A step in a retrieval pipeline where an initial set of candidate results is reordered by a more precise (often more computationally expensive) model to improve the relevance of the top results.

Named Entity Recognition (NER)

An NLP task that identifies and classifies named entities in text, such as people, organizations, dates, and locations.

Sentiment Analysis

An NLP task that determines the emotional tone or attitude expressed in a piece of text, such as classifying it as positive, negative, or neutral.

Word Embedding

A vector representation of a word that captures aspects of its meaning based on the contexts in which it appears, such that semantically similar words have similar vectors.

BLEU Score

A metric used to evaluate the quality of machine-translated or generated text by comparing overlapping word sequences ('n-grams') with one or more reference texts.

Perplexity

A metric for evaluating language models that measures how well a probability model predicts a sample; lower perplexity indicates the model is less 'surprised' by the actual text.

Training, Optimization & Model Behavior

Optimizer

An algorithm used to update a model's parameters during training to minimize the loss function; common examples include SGD, Adam, and RMSprop.

Stochastic Gradient Descent (SGD)

A variant of gradient descent that updates model parameters using a small random subset (mini-batch) of the training data at each step, rather than the entire dataset.

Adam Optimizer

A widely used optimization algorithm that adapts the learning rate for each parameter individually, combining ideas from momentum and adaptive learning rate methods.

Transfer Learning

A technique where a model trained on one task is reused, often with fine-tuning, as the starting point for a related task, reducing the data and compute needed for the new task.

Knowledge Distillation

A technique for training a smaller 'student' model to replicate the behavior of a larger 'teacher' model, achieving similar performance with reduced computational cost.

Quantization

A technique that reduces the numerical precision of a model's weights (for example, from 32-bit to 8-bit numbers) to shrink its size and speed up inference, typically with some tradeoff in accuracy.

Pruning

A technique that removes less important weights or neurons from a trained neural network to reduce its size and computational cost.

Inference

The process of using a trained model to make predictions or generate outputs on new input data, as opposed to training the model.

Latent Space

An internal, often lower-dimensional, representation space learned by a model where similar inputs are positioned closer together, capturing underlying structure in the data.

Generative Adversarial Network (GAN)

A model architecture consisting of two neural networks, a generator and a discriminator, trained together in competition so the generator learns to produce increasingly realistic synthetic data.

Diffusion Model

A generative model that learns to create data, such as images, by starting from random noise and iteratively refining it into a coherent output through a learned denoising process.

Autoencoder

A neural network trained to compress input data into a compact representation and then reconstruct the original input from it, often used for dimensionality reduction or anomaly detection.

Catastrophic Forgetting

A phenomenon where a neural network loses previously learned information after being trained on new data or tasks.

Model Evaluation & Metrics

Accuracy

The proportion of predictions a model gets correct out of all predictions made; a simple but sometimes misleading metric on imbalanced datasets.

Precision

Of all the instances a model predicted as positive, the proportion that were actually positive; a measure of how trustworthy positive predictions are.

Recall

Of all the instances that were actually positive, the proportion the model correctly identified; a measure of how completely the model finds positive cases.

F1 Score

A single metric that balances precision and recall by taking their harmonic mean, useful when both false positives and false negatives matter.

Confusion Matrix

A table summarizing a classification model's performance by showing counts of true positives, true negatives, false positives, and false negatives.

ROC Curve

A graph that shows the tradeoff between a classification model's true positive rate and false positive rate across different decision thresholds.

AUC (Area Under the Curve)

A single number summarizing a model's ROC curve, representing the probability that the model ranks a randomly chosen positive example higher than a randomly chosen negative one.

Mean Squared Error (MSE)

A common loss and evaluation metric for regression tasks that averages the squared differences between predicted and actual values, penalizing larger errors more heavily.

A/B Testing

An experimental method for comparing two versions of a system by randomly showing each version to different user groups and measuring which performs better on a chosen metric.

Benchmark

A standardized dataset or task used to evaluate and compare the performance of different models under consistent conditions.

MLOps, Deployment & Systems

MLOps

A set of practices for deploying, monitoring, and maintaining machine learning models reliably and efficiently in production, borrowing ideas from DevOps.

Model Drift

A decline in a deployed model's performance over time as the statistical properties of real-world input data change from what the model was originally trained on.

Data Pipeline

A series of automated steps that move and transform raw data into a form suitable for training or serving a machine learning model.

API (Application Programming Interface)

A defined set of rules and protocols that allows different software systems, such as an application and a machine learning model, to communicate with each other.

Latency

The time delay between sending a request to a system, such as a model inference request, and receiving a response.

Throughput

The number of requests or operations a system, such as a model serving pipeline, can process in a given period of time.

GPU (Graphics Processing Unit)

A specialized processor originally designed for rendering graphics, whose ability to perform many parallel calculations makes it well suited to training and running neural networks.

Model Checkpoint

A saved snapshot of a model's parameters at a particular point during training, allowing training to resume or the model to be evaluated or deployed later.

AI Ethics & Safety

Algorithmic Bias

Systematic and unfair discrimination in a model's outputs, often arising from biased training data, flawed assumptions, or unrepresentative sampling.

Explainability (XAI)

The degree to which a model's internal decision-making process can be understood and articulated in terms humans can interpret.

Interpretability

The extent to which a human can understand the cause of a model's decision or prediction, often used alongside explainability.

Alignment

The broad goal and set of techniques aimed at ensuring an AI system's behavior and objectives match the intentions and values of its human developers and users.

Adversarial Example

An input deliberately crafted with small, often imperceptible, perturbations designed to cause a machine learning model to make an incorrect prediction.

Differential Privacy

A mathematical framework for training models or releasing data statistics in a way that limits how much information about any single individual can be inferred from the output.

Federated Learning

A training approach where a model is trained across many decentralized devices or servers holding local data, without that raw data ever leaving its original location.

Guardrails

Rules, filters, or constraints placed around an AI system's inputs or outputs to prevent unsafe, inappropriate, or unintended behavior.